线性结构

1. B. 2. B. 3. A 4. C. 5. C.

6. a_4 a_3 a_2 a_1

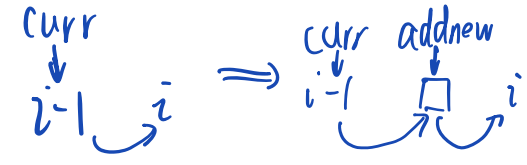
7. B. 8. C. 9. D. 10. A 11. C.

12. D. B. C. A

13. A. 14. B. 15. D. 16. A 17. A.

18. (1) add (int x, int i):

```
S1   if (i == 0){ addnew.next = first;        → 在头部插入
                  first = addnew; }
S2   curr = first; index = 0;
S3   while (curr.next!=null && index < i-1)
     {   curr = curr.next;                    → 遍历至 i-1
         index ++; }
S4   addnew.next = curr.next;
     curr.next = addnew;
```

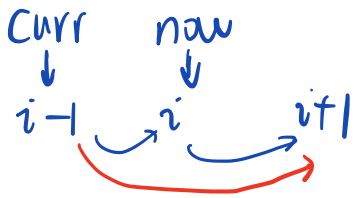


(2) remove (int i):

```
S1   if (first == null){ return; }
     if (i == 0){ first = first.next; }
S2   curr = first; index = 0;
S3   while (curr.next!=null && index < i-1):
     {  curr = curr.next; index++; }
```

S4 now = curr.next;

curr.next = now.next;



树

1. 176

2. 叶结点：⑨⑧⑥

分支结点：①②③④⑤⑦

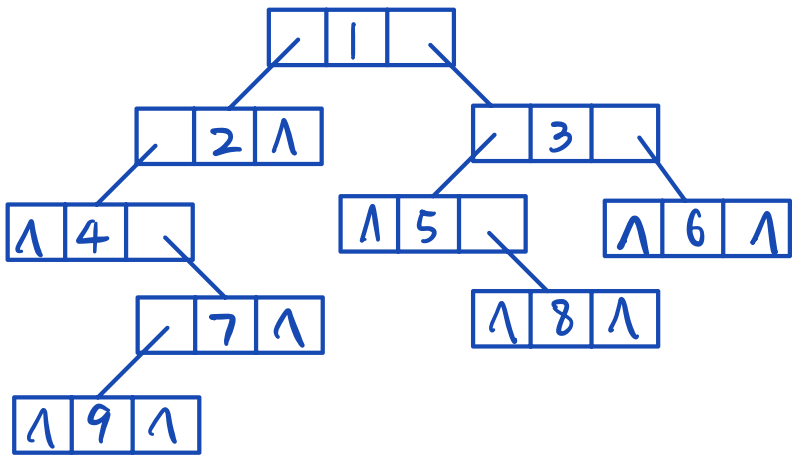
层次：①：0　②③：1　④⑤⑥：2

⑦⑧：3　⑨：4

3. (1)

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
data	1	2	3	4		5	6		7				8					9		

(2)

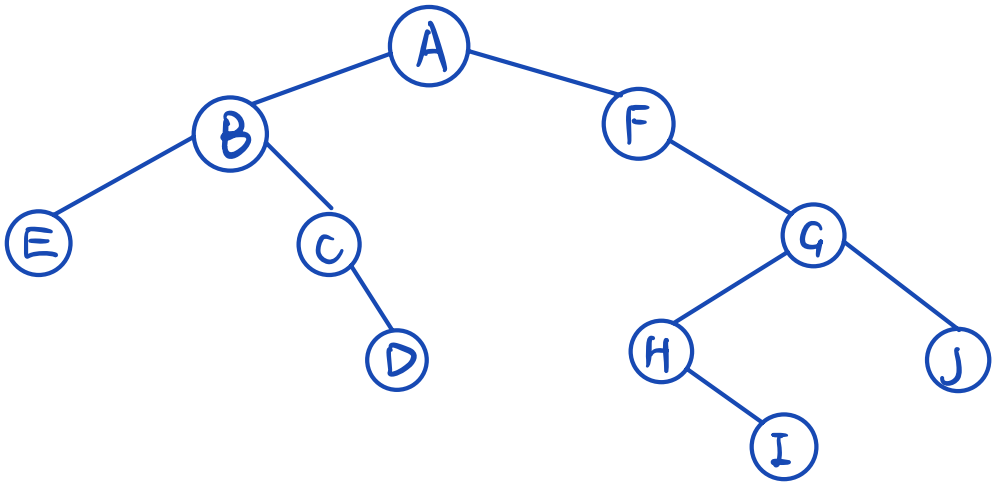


4. 先序遍历：12479358 6

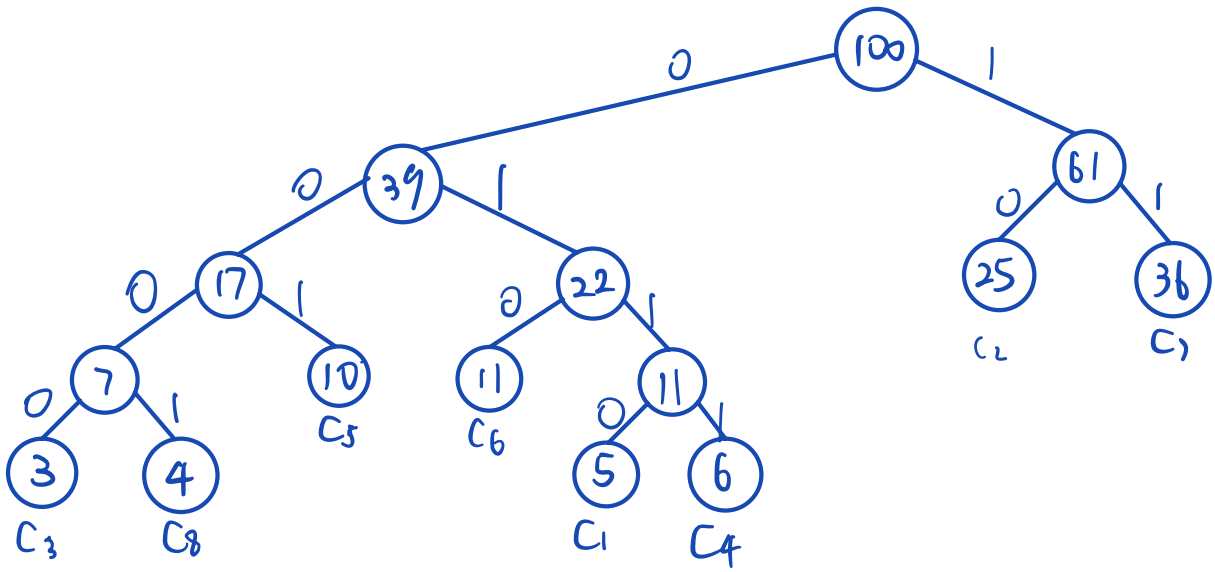
中序遍历：49721583 6

后序遍历：97428 5631

5.



6.



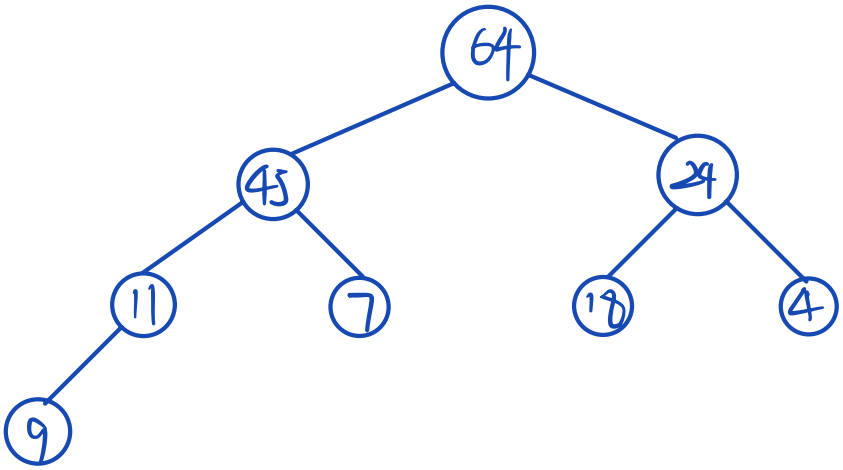
∴

C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
0110	10	0000	0111	001	010	11	0001

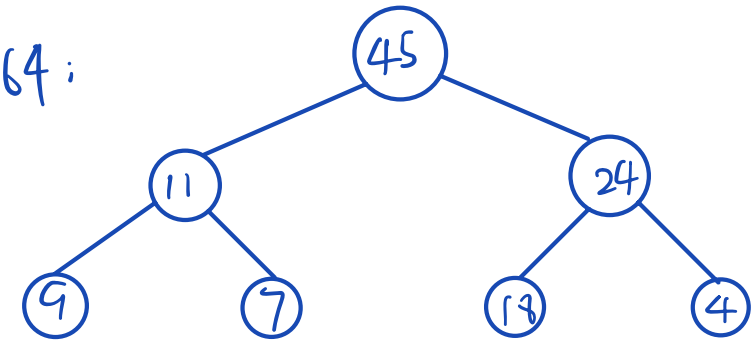
总码数：$4\times5+2\times25+4\times3+4\times6+3\times10+3\times11+2\times36+4\times4$

$=257$

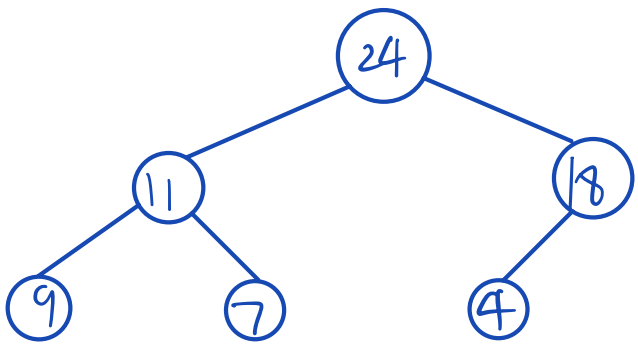
7. 不是。



输出64：

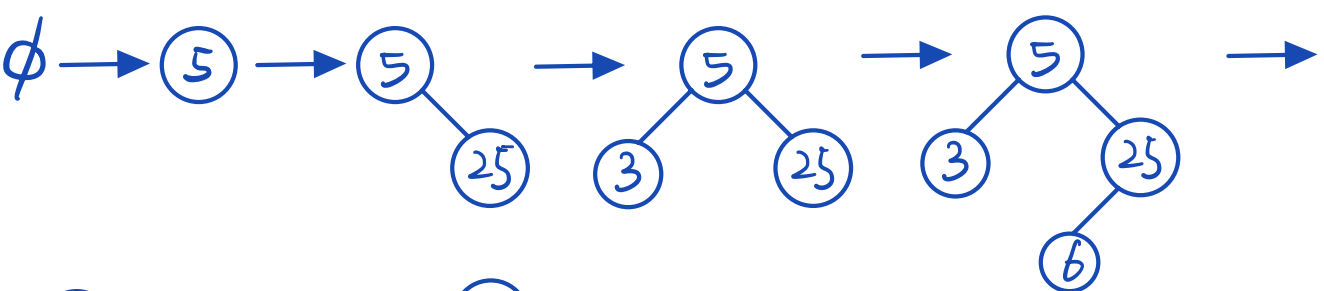


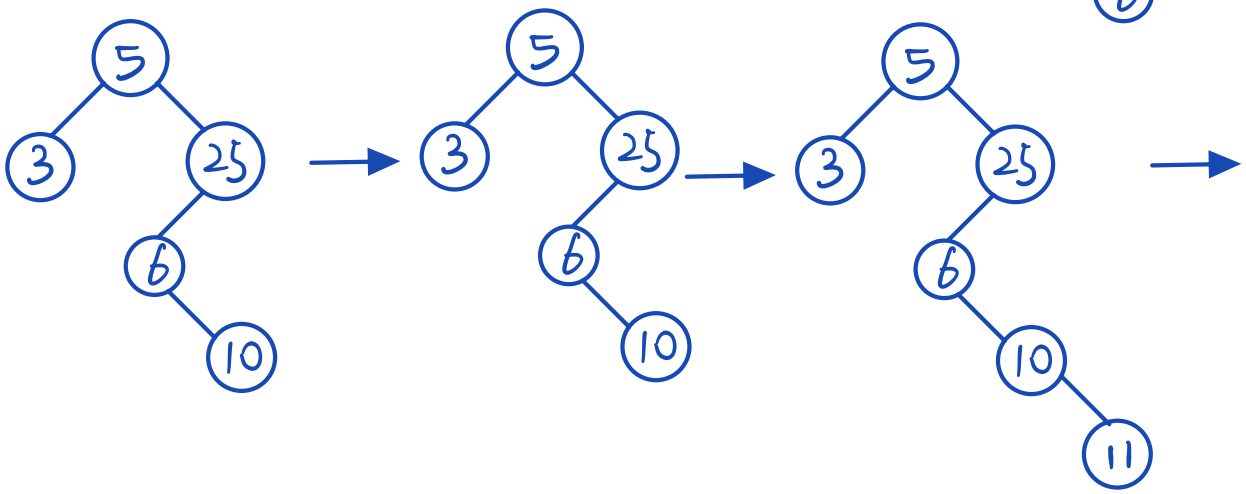
输出45：

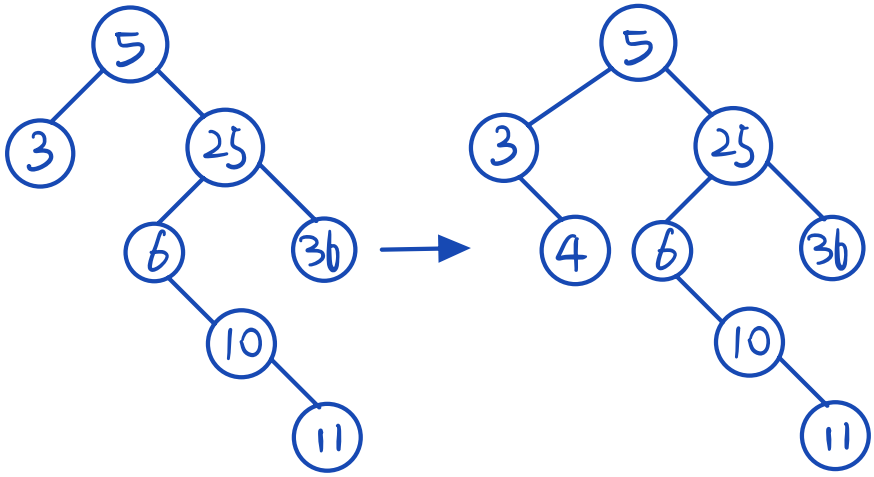


8. (1) ϕ → 5 → 5 → 5 → 5 →

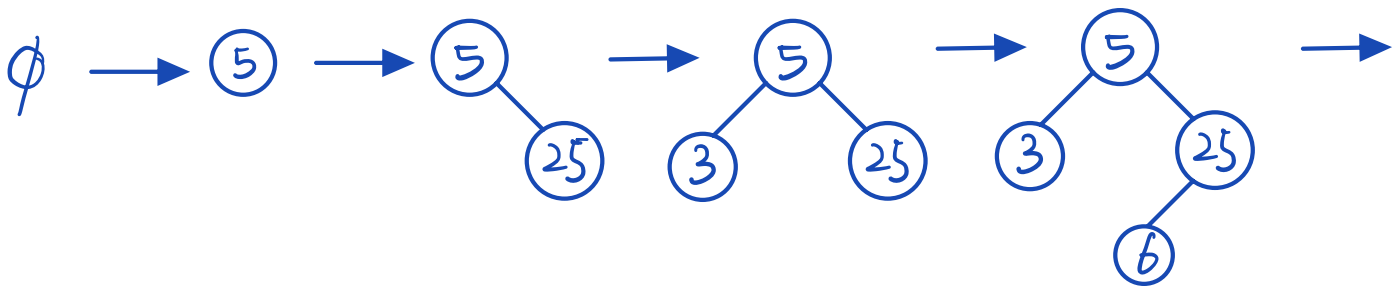
BST：

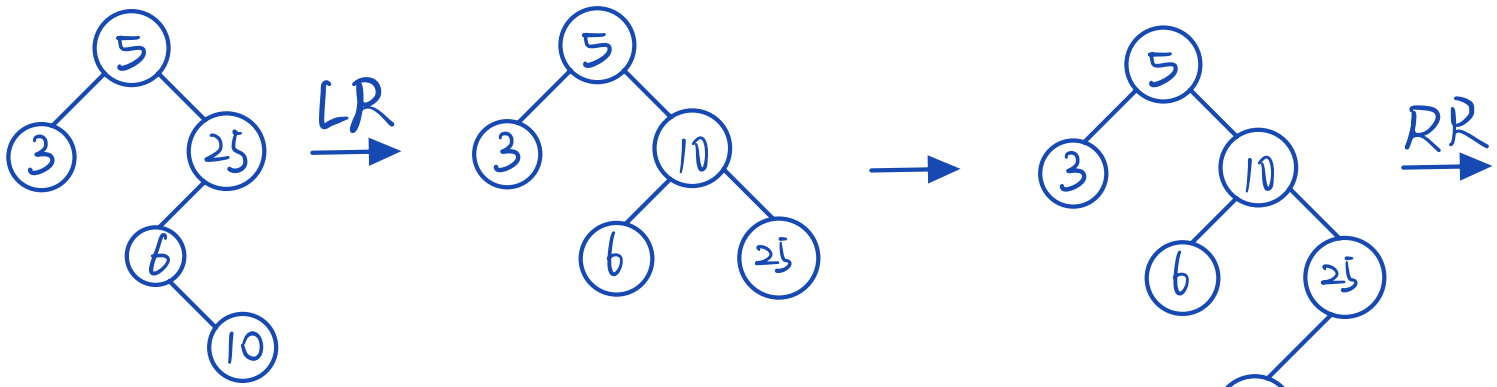


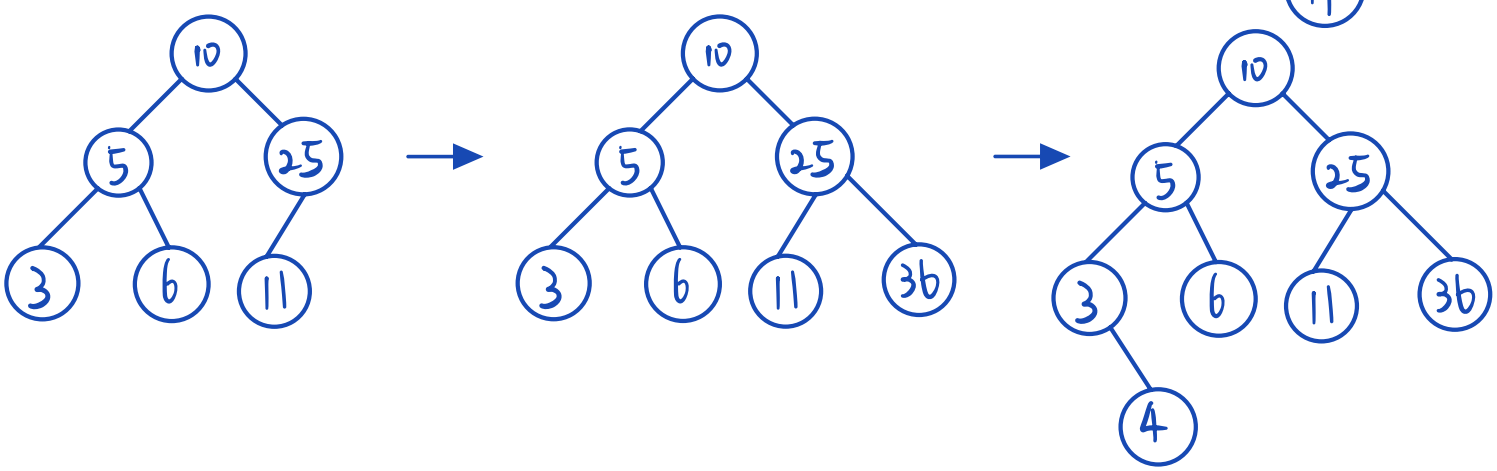



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(2) AVL :


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9. (1)
```
int countLeaves(TreeNode root):
S1   if (root == null) { return 0; }
S2   if (root.left == null && root.right == null)
       { return 1; }
S3   return countLeaves(root.left) + countLeaves(root.right)
```

(2)
```
void swapChildren(TreeNode root):
S1   if (root == null) { return 0; }
S2   TreeNode temp = root.left;
     root.left = root.right;
     root.right = temp;
S3   swapChildren(root.left);
     swapChildren(root.right);
```